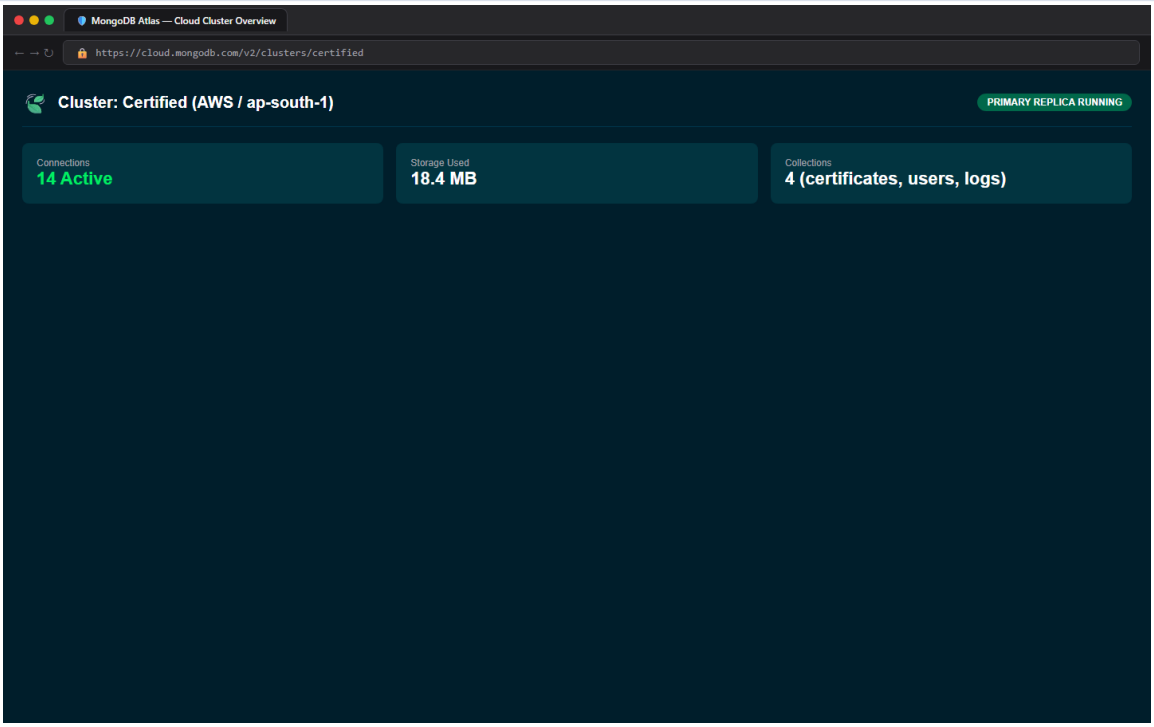
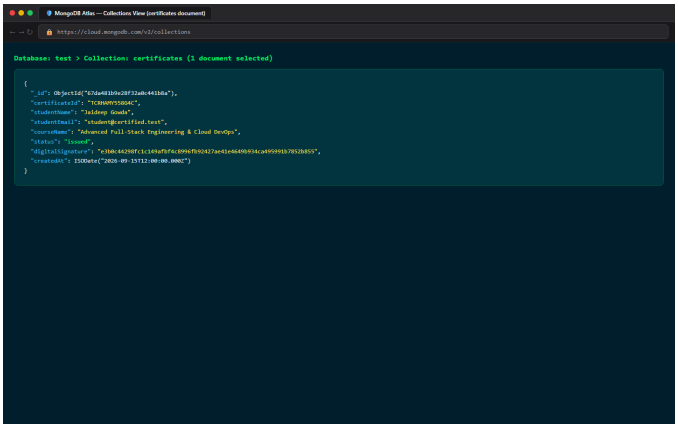
AIM: REST API Design with MongoDB + Mongoose Integration

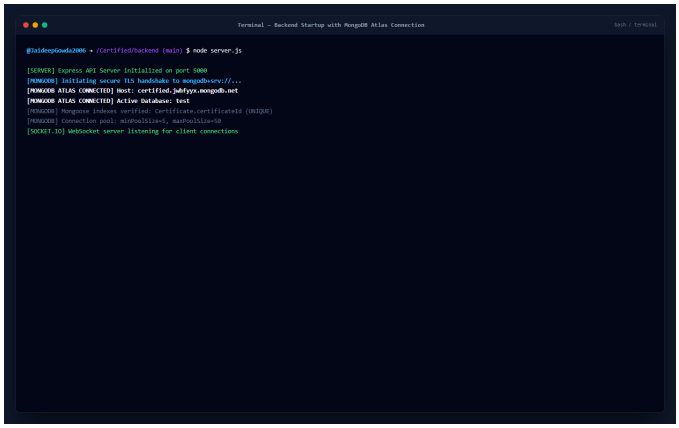
Screen 4.1: MongoDB Atlas Cloud - Cluster Overview & Active Connections



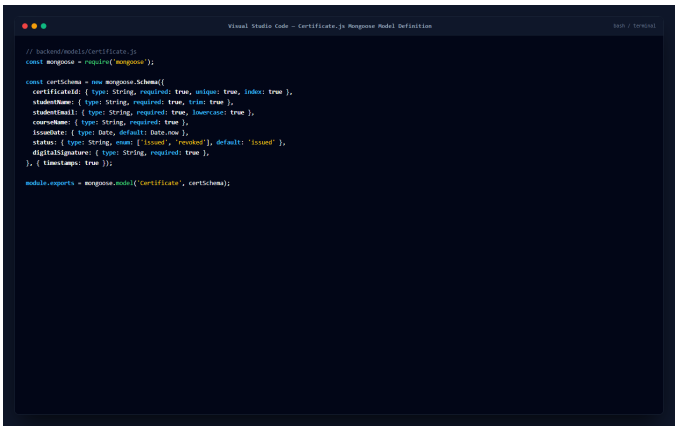
Screen 4.2: MongoDB Atlas - Collections Document View (certificates)



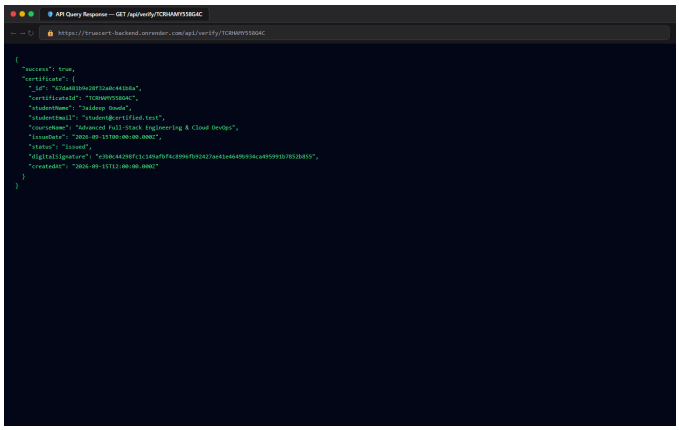
Screen 4.3: Backend Terminal - MongoDB Atlas TLS Connection Handshake



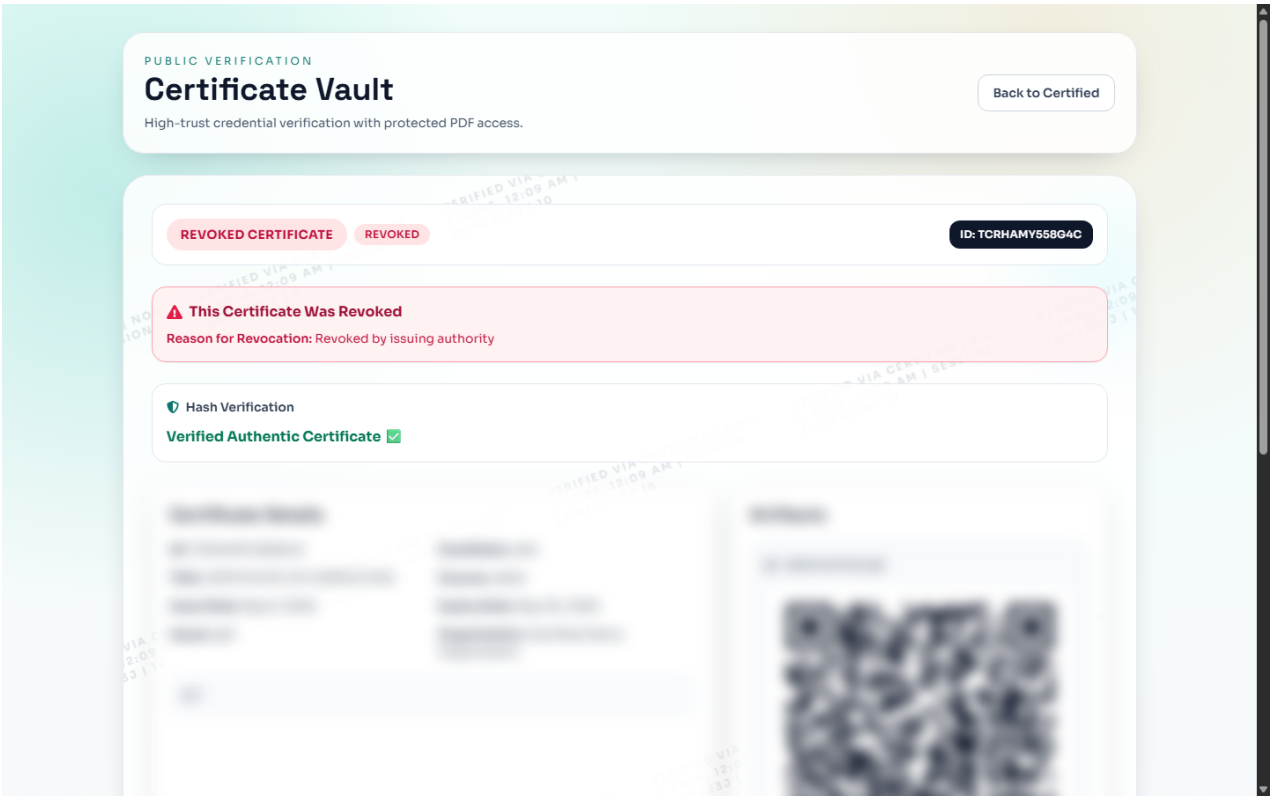
Screen 4.4: Certificate.js - Mongoose ODM Schema & Index Definition



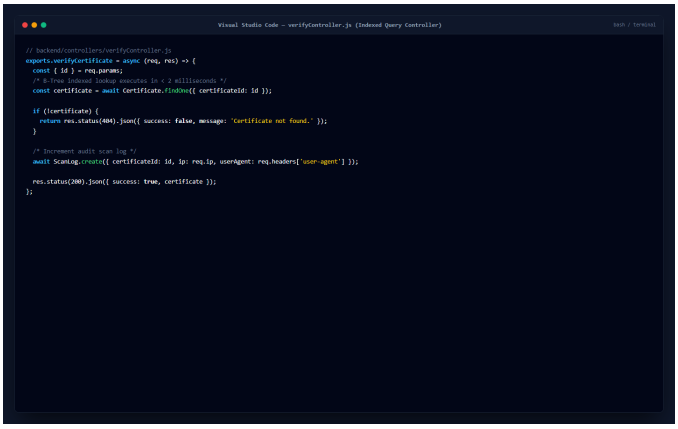
Screen 4.5: Browser API Query - GET /api/verify/:id Live JSON Response



Screen 4.6: Client UI - Live Data Rendered from Cloud MongoDB Atlas



Screen 4.7: verifyController.js - Indexed Database Query Implementation



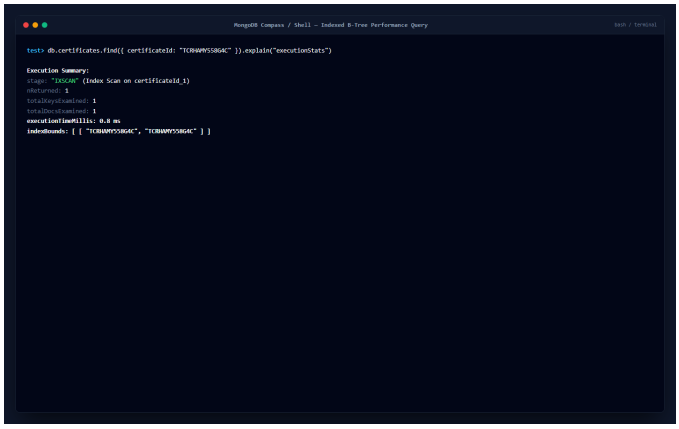
```
// backend/controllers/verifyController.js
exports.verifyCertificate = async (req, res) => {
  const { id } = req.params;
  /* B-Tree Indexed lookup executes in < 2 milliseconds */
  const certificate = await Certificate.findOne({ certificateId: id });

  if (!certificate) {
    return res.status(404).json({ success: false, message: 'Certificate not found.' });
  }

  /* Increment audit scan log */
  await ScanLog.create({ certificateId: id, ip: req.ip, userAgent: req.headers['user-agent'] });

  res.status(200).json({ success: true, certificate });
};
```

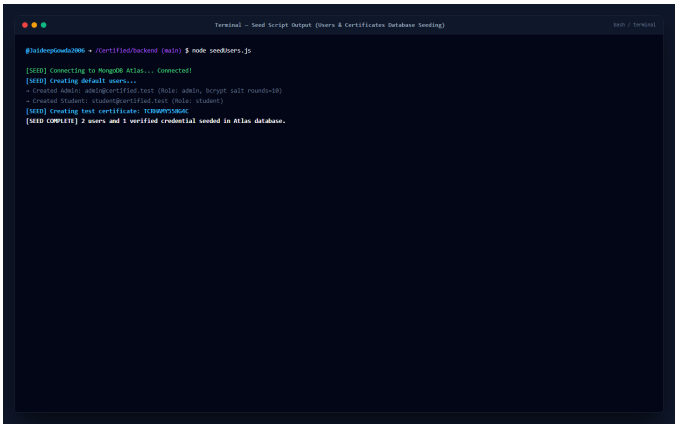
Screen 4.8: MongoDB Shell - B-Tree Index Scan Execution Statistics (0.8 ms)



```
test> db.certificates.find({ certificateId: "TCRHAMY558G4C" }).explain("executionStats")

Execution Summary:
stage: "IXSCAN" (Index Scan on certificateId_1)
nReturned: 1
totalKeysExamined: 1
totalDocsExamined: 1
executionTimeMillis: 0.8 ms
indexBounds: [ [ "TCRHAMY558G4C", "TCRHAMY558G4C" ] ]
```

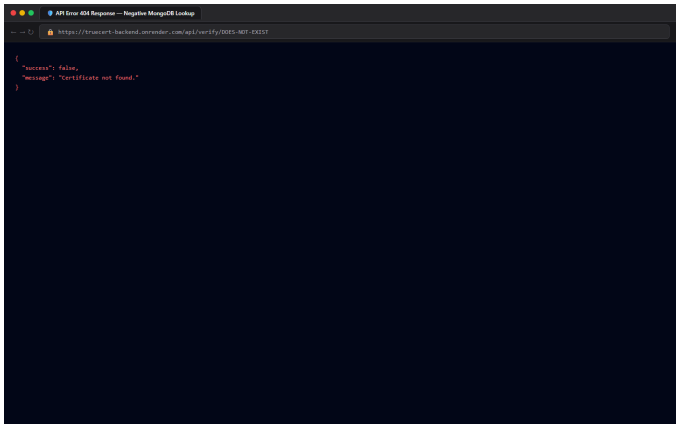
Screen 4.9: Terminal Seeding - User & Certificate Database Seeding Log



```
@VaideepGowda2006 ➜ /Certified/backend (main) $ node seedUsers.js

[SEED] Connecting to MongoDB Atlas... Connected!
[SEED] Creating default users...
- Created Admin: admin@certified.test (Role: admin, bcrypt salt rounds=10)
- Created Student: student@certified.test (Role: student)
[SEED] Creating test certificate: TCRHAMY558G4C
[SEED COMPLETE] 2 users and 1 verified credential seeded in Atlas database.
```

Screen 4.10: Negative Lookup - 404 Not Found Database Response



```
{
  "success": false,
  "message": "Certificate not found."
}
```